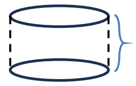
Section B

8 **(a)** **(i)** Area cut in time ⊗*t* is the curved surface area of a cylinder traced out by the falling ring. $A = (2\pi r)(v\Delta t)$

Flux cut, $\Delta\Phi = BA = B(2\pi r)(v\Delta t)$ B1



Comments: Some explanation is expected in the working since this is a "show" question. No credit for candidates who just write $\Delta\Phi = B(2\pi r)(v\Delta t)$

(ii) From Faraday's Law, induced e.m.f. $E = \dfrac{\Delta\Phi}{\Delta t}$

$$= \frac{B(2\pi r)(v\Delta t)}{\Delta t} = 2\pi rBv$$ M1

Induced current $I = \dfrac{E}{R}$ M1

$$= \frac{2\pi rBv}{R}$$

(iii) Magnetic force exerted by the radial magnetic field on the induced current in the ring, $F_B = BIL = B(\quad)(2\pi r)$ M1

From Newton's 2nd Law: Resultant force on the ring $= mg - F_B = ma$ M1

$$mg - B\left(\frac{2\pi rBv}{R}\right)(2\pi r) = ma$$

$$a = g - \frac{(2\pi rB)^2 v}{mR}$$ A0

(iii) Maximum speed (when a = 0) is

$$v_{max} = \frac{mgR}{(2\pi rB)^2}$$ C1

$$= \frac{(0.0235)(9.81)(2.30\times10^{-4})}{(2\pi\times0.03\times0.800)^2}$$ M1

$= 2.33 \times 10^{-3}$ m s^{-1} A1

(iv)

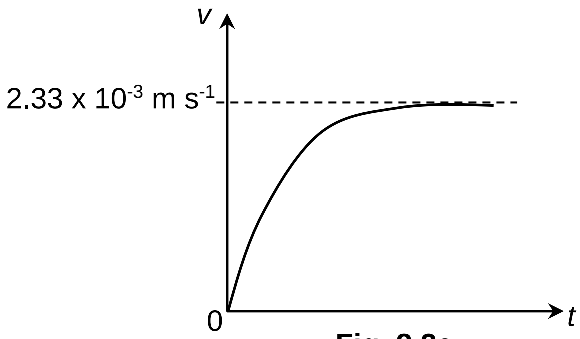


Fig. 8.2a

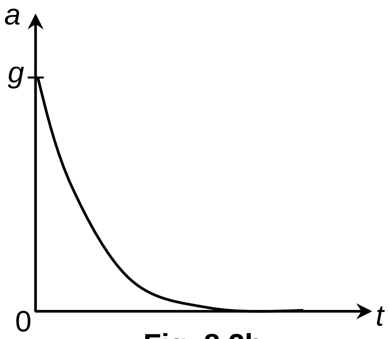


Fig. 8.2b

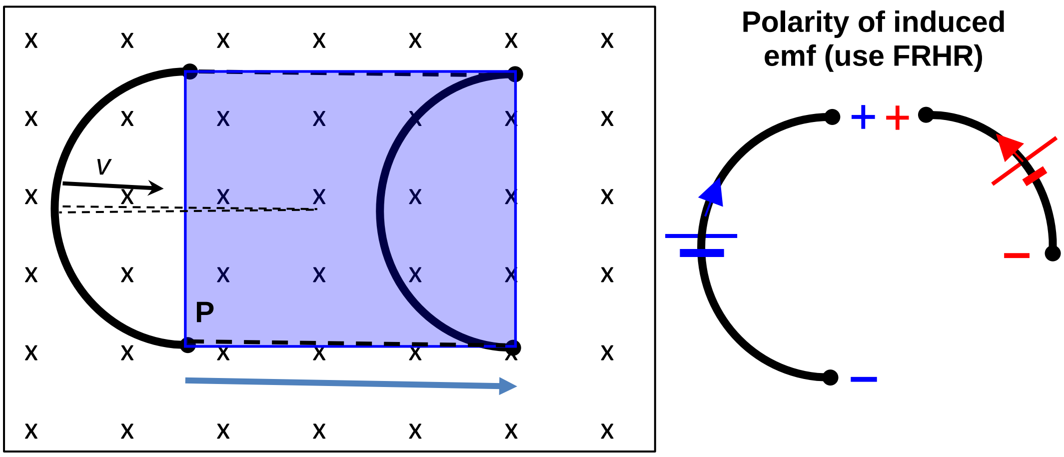
Correct shape	B1
label terminal velocity	B1
Correct shape for a-t graph	B1

(b) **(i)** Induced e.m.f. = *Brv* = 0.500 x 0.03 x 0.03 — M1

= 4.50 x 10^{-4} V — A1

(ii) Q — B1

Explanation for (i) above



Area cut/swept in time t is equal to the area of the shaded rectangle.

Induced emf for the semi-circle.

Induced emf for the quarter circle

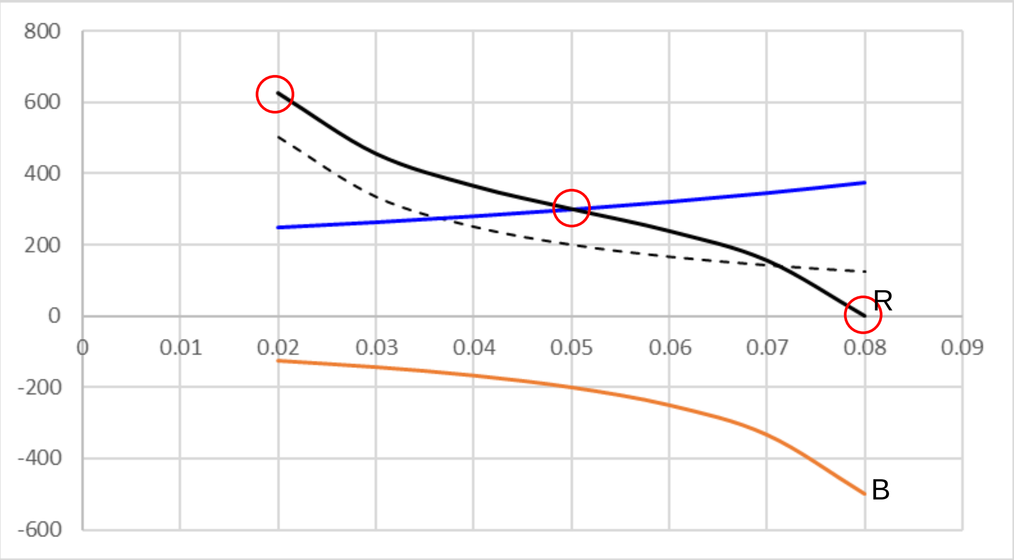
Net induced emf

(c) **(i)** Flux density due to the two conductors carrying 20 A will cancel at X leaving the resultant flux density = $\frac{\mu_0(90)}{2\pi\times 0.15}$ — M1

= 1.20 x 10^{-4} T — A1

Out of the page — B1

(ii)



Correct shape and negative values for curve B B1

Correct shape for R B1

R must pass through 2 out of 3 circled points. B1

9 **(a)** **(i)** lowest frequency of electromagnetic radiation M1